Math 125 - Winter 2015
Exam 2
February 26, 2015

Name: _____

Section: _____

Student ID Number: _____

PAGE 1	12	
PAGE 2	12	
PAGE 3	13	
PAGE 4	13	
PAGE 5	10	
Total	60	

- There are 5 pages of questions. Make sure your exam contains all these questions.
- No calculators allowed!
- You are allowed one **hand-written** 8.5 by 11 inch page of notes.
- You must show your work on all problems. The correct answer with no supporting work may result in no credit. **Put a box around your FINAL ANSWER for each problem and cross out any work that you don't want to be graded.** Give exact answers wherever possible.
- If you need more room, use the backs of the pages and indicate to the grader that you have done so.
- Raise your hand if you have a question.
- There may be multiple versions of the exam so if you copy off a neighbor and put down the answers from another version we will know you cheated. Any student found engaging in academic misconduct will receive a score of 0 on this exam. All suspicious behavior will be reported to the student misconduct board. In such an instance, you will be force to meet in front of a board of professors to explain your actions.
  DO NOT CHEAT OR DO ANYTHING THAT LOOKS SUSPICIOUS!
  WE WILL REPORT YOU AND YOU MAY BE EXPELLED!
- You have 80 minutes to complete the exam. Budget your time wisely.
  SPEND NO MORE THAN 15 MINUTES PER PAGE!

GOOD LUCK!

1. (12 points) Compute:

   (a) $\int \tan^{-1}(x)\, dx.$

   (b) $\int \dfrac{x^2 - 6}{2x^2 - x^3}\, dx.$

2. (12 points) Compute:

(a) $\displaystyle\int \frac{x}{(x^2+4x+13)^{3/2}}\,dx.$

(b) $\displaystyle\int_0^{\pi/2} \cos^4(x)\sin^3(x)\,dx$

3. (13 points)

(a) Compute: $\displaystyle\int \frac{\sqrt{x}}{\sqrt{x}+3}\,dx.$

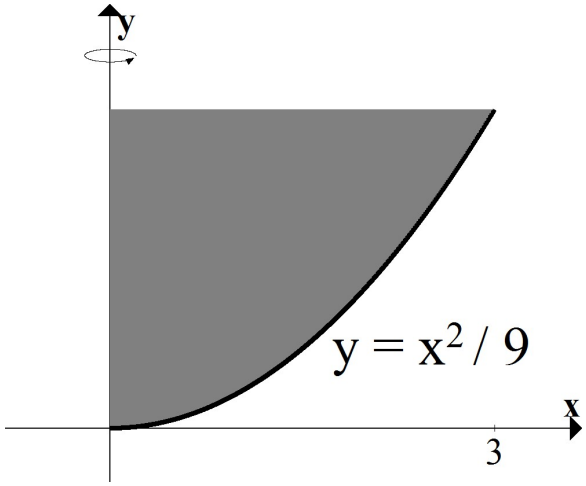
(b) Consider the arc length of the curve $f(x) = 30\ln(x)$ from $x = 1$ to $x = 3$.

i. (2 pts) Set up (DO NOT EVALUATE) an integral for this arc length.

ii. (5 pts) Use Simpson's rule with $n = 4$ to approximate the value of the arc length. (You do not need to simplify your answer, leave it expanded out)

4. (13 points)

(a) (7 pts) Evaluate the following **improper** integral: $\displaystyle\int_0^\infty \frac{\sin(\pi e^{-x})}{e^x}\,dx$.
(Give the value if it converges, or show why it diverges).

(b) The temperature for a particular object after t minutes is given by $T(t) = 4te^{-2t}$ degrees Celsius. Find the average temperature from $t = 0$ to $t = 3$ minutes.

5. (10 points) The portion of the graph $y = \frac{1}{9}x^2$ between $x = 0$ and $x = 3$ is rotated around the y-axis to form a container. The container is full of a liquid that has density 100 lbs/ft^3.
Find the work required to pump all of the liquid out over the side of the container.
(Distance is measured in feet).